MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

1. A rectangle is formed by lines $x = 0$, $y = 0$, $x = 3$, $y = 4$. A line perpendicular to $3x + 4y + 6 = 0$ divides the rectangle into two equal parts, then the distance of the line from $\left(-1, \frac{3}{2}\right)$ is

(1) 2 (2) $\frac{17}{10}$

(3) $\frac{6}{5}$ (4) $\frac{8}{5}$

Answer (2)

Sol. Concept : line passes through centre of the rectangle $\equiv \left(\frac{3}{2}, 2\right)$

$\Rightarrow 4x - 3y = C$ passes through $\left(\frac{3}{2}, 2\right)$

$\Rightarrow C = 0$

$\Rightarrow$ distance of $\left(-1, \frac{3}{2}\right)$ from $4x - 3y = 0$ is $\frac{\left|-4 - \frac{9}{2}\right|}{5} = \frac{17}{10}$

2. Let $A = \{-2, -1, 0, 1, 2, 3, 4\}$ and R be a relation R, such that $R = \{(x, y) : (2x + y) \le -2, x \in A, y \in A\}$

Let l = number of elements in R

m = minimum number of elements to be added in R to make it reflexive

n = minimum number of elements to be added in R to make it symmetric, then $(l + m + n)$ is

(1) 10 (2) 17

(3) 11 (4) 14

Answer (2)

Sol. $R = \{(x, y) : 2x + y \le -2, x \in A, y \in A\}$

$A = \{-2, -1, 0, 1, 2, 3, 4\}$

$R = \{(-2, 2), (-2, 1), (-2, 0), (-2, -1), (-2, -2), (-1, 0), (-1, -1), (-1, -2), (0, -2)\}$

$l = 9$

$m = 5$

$n = \{(-2, 1), (-2, 2), (-1, 0)\} = 3$

3. Number of 4 letter words with or without meaning formed from the letters of the word PQRSSSTTUVV is

(1) 1232 (2) 1400

(3) 1422 (4) 1162

Answer (3)

Sol. Case A : The word is ABCD

$\Rightarrow {}^7C_4 \cdot 4! = 840$

Case B: The word is AAAB

$\Rightarrow (SSS) \times ({}^6C_1) \Rightarrow {}^6C_1 \cdot \frac{4!}{3!} = 24$

Case C : The word is AABB

$\Rightarrow {}^3C_2 \cdot \frac{4!}{2!2!} = 18$

Case (D) : The word is AABC

$\Rightarrow \left({}^3C_1\right) \cdot \left({}^6C_2\right) \cdot \frac{4!}{2!} = 3 \times 15 \times 12 = 540$

Answer 1422

4. The area (in square units) bounded by the curve $f(x) = \max\{\sin x, \cos x\}$ and x-axis between $x = 0$ and $x = \frac{3\pi}{2}$ is A sq. unit. Then, the value of $A + A^2$ is

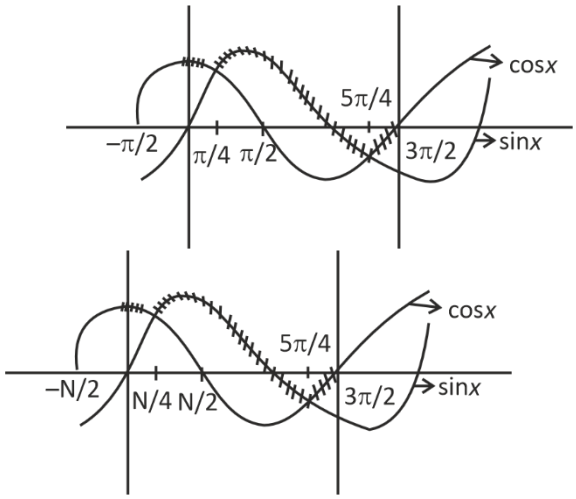
(1) $8 - 2\sqrt{2}$ (2) $8 + 2\sqrt{2}$

(3) $6 - 2\sqrt{2}$ (4) $6 + 2\sqrt{2}$

Answer (1)



Sol.



$$\text{Required Area} = = \int_0^{\pi/4} \cos x dx + \int_{\pi/4}^{5\pi/4} \sin x dx$$

$$+ \int_{5\pi/4}^{3\pi/2} \cos x dx$$

$$= [\sin x]_0^{\pi/4} + [-\cos x]_{\pi/4}^{5\pi/4} + [\sin x]_{5\pi/4}^{3\pi/2}$$

$$= \frac{1}{\sqrt{2}} - \left[\frac{-1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right] + \left[-1 + \frac{1}{\sqrt{2}}\right]$$

$$= \frac{1}{\sqrt{2}} + \sqrt{2} - 1 + \frac{1}{\sqrt{2}}$$

$$A = 2\sqrt{2} - 1$$

$$A + A^2 = (2\sqrt{2} - 1) + (2\sqrt{2} - 1)^2$$

$$= 2\sqrt{2} - 1 + 8 + 1 - 4\sqrt{2} = 8 - 2\sqrt{2} \text{ sq. unit}$$

5. The value of $\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \ldots\ldots \frac{{}^{100}C_{100}}{101}$ is

(1) $\frac{2^{100}}{100}$ (2) $\frac{2^{101}}{101}$

(3) $\frac{2^{100}}{101}$ (4) $\frac{2^{101}}{100}$

Answer (3)

Sol. $(1+x)^{100} = {}^{100}C_0 + {}^{100}C_1 x + {}^{100}C_2 x^2 + \ldots\ldots + {}^{100}C_{100} x^{100}$

$$\sum_{r=50}^{100} \frac{{}^{100}C_r}{r+1} = \frac{1}{101} \sum_{r=50}^{100} {}^{101}C_{r+1}$$

$$\Rightarrow \frac{1}{101} \sum_{r=51}^{100} {}^{101}C_r$$

$$\Rightarrow \frac{2^{100}}{101}$$

6. For given vectors $\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ where $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{b}$. Then the value of $(\vec{a} - \vec{b}) \cdot \vec{d}$ is

(1) -35 (2) 53

(3) -52 (4) 25

Answer (1)

Sol. $\vec{a} = -\hat{c} + \hat{j} + 2\hat{k}$

$\vec{b} = 2\hat{c} - \hat{j} + \hat{k}$

$\vec{c} = \vec{a} \times \vec{b} = 3\hat{c} + 5\hat{j} - \hat{k}$

$\vec{d} = \vec{c} \times \vec{b}$

Now

$$(\vec{a} - \vec{b}) \cdot \vec{d} = (\vec{a} - \vec{b}) \cdot (\vec{c} \times \vec{b})$$

$$= \vec{a} \cdot (\vec{c} \times \vec{b}) - \vec{b} \cdot (\vec{c} \times \vec{b})$$

$$= -[\vec{a}\ \vec{b}\ \vec{c}]$$

$$= -35$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{c} & \hat{j} & \hat{k} \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix} = 3\hat{c} + 5\hat{j} - \hat{k}$$

And $[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} -1 & 1 & 2 \\ 2 & -1 & 1 \\ 3 & 5 & -1 \end{vmatrix} = 35$

$\Rightarrow \quad -[\vec{a}\ \vec{b}\ \vec{c}] = -35$



7. The line $y = x + 1$ intersects the ellipse $\frac{x^2}{2}+\frac{y^2}{1}=1$ at A and B. Find the angle sub-stained by segment AB and centre of ellipse is

(1) $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$

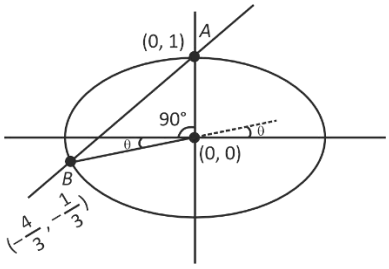
(2) $\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{4}\right)$

(3) $\frac{\pi}{2}+2\tan^{-1}\left(\frac{1}{4}\right)$

(4) $\frac{\pi}{4}+\tan^{-1}\left(\frac{1}{4}\right)$

Answer (1)

Sol.



$y = x + 1$ intersects $\frac{x^2}{2}+y^2=1$

$\Rightarrow \frac{x^2}{2}+(x+1)^2=1 \Rightarrow x^2 + 2x^2 + 4x = 0$

$\Rightarrow x = 0, \frac{-4}{3}$

$\Rightarrow$ Points A and B are

$(0, 1), \left(\frac{-4}{3},\frac{-1}{3}\right)$

$\theta=\tan^{-1}\left(\frac{\frac{-1}{3}}{\frac{-4}{3}}\right)=\tan^{-1}\left(\frac{1}{4}\right)$

$\Rightarrow \angle AOB = \frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$

8. Find $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}}\frac{dx}{1+(\tan 2x)^{1/3}}$

(1) $\frac{\pi}{24}$ (2) $\frac{\pi}{12}$

(3) $\frac{\pi}{6}$ (4) $\frac{\pi}{48}$

Answer (2)

Sol. $I=\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}}\frac{1}{1+(\tan 2x)^{1/3}}dx$ ...(1)

$$\Rightarrow I=\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}}\frac{1}{1+\tan\left(2\left(\frac{5\pi}{24}+\frac{\pi}{24}-4\right)\right)^{1/3}}dx$$

$$I=\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}}\frac{1}{1+\tan\left(\frac{\pi}{2}-2x\right)^{1/3}}dx \quad ...(2)$$

From equation (1) and (2)

$$2I=\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}}\frac{(\tan(2x))^{1/3}+1}{1+(\tan 2x)^{1/3}}dx$$

$$=(x)\Big|_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \Rightarrow I=\frac{1}{2}\times\frac{4\pi}{24}=\frac{\pi}{12}$$

9. The number of solutions of $13\cos 2\theta+8\cos\theta-3\sqrt{3}=0$ if $\theta\in[-3\pi,2\pi]$ is

(1) 4 (2) 3

(3) 0 (4) 5

Answer (4)

Sol. $13[2\cos^2\theta-1]+8\cos\theta-3\sqrt{3}=0$

$26\cos^2\theta+8\cos\theta-13-3\sqrt{3}=0$

$$\cos\theta=\frac{-8\pm\sqrt{64+4\times26(13+3\sqrt{3})}}{52}$$

$$=\frac{-8\pm\sqrt{1416+312\sqrt{3}}}{52}\approx 0.69$$

$\therefore$ 5 Solution



10. For the differential equation $x^4dy + (4x^3y + \sin x)dx = 0$ it is given that $y\left(\frac{\pi}{2}\right)=0$. Then the value of $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to

(1) $\frac{9}{2}$ (2) $\frac{81}{2}$

(3) $\frac{27}{2}$ (4) $\frac{243}{2}$

Answer (2)

Sol. $x^4dy + (4x^3y + \sin x)dx = 0$

$$\frac{dy}{dx}+\frac{4x^3}{x^4}y=\frac{-\sin x}{x^4}$$

$$\frac{dy}{dx}+\frac{4}{x}y=\frac{-\sin x}{x^4}$$

IF = $e^{\int\frac{4}{x}dx}=e^{\ln x^4}=x^4$

$$x^4\frac{dy}{dx}+4x^3y=-\sin x$$

$$\frac{d}{dx}(x^4y)=-\sin x$$

$\Rightarrow x^4 + y = \cos x + C$

$$y\left(\frac{\pi}{2}\right)=0$$

$\Rightarrow C = 0$

$$\therefore\ x^4y=\cos x\Rightarrow y=\frac{\cos x}{x^4}$$

$$y\left(\frac{\pi}{3}\right)=\frac{\cos\left(\frac{\pi}{3}\right)}{\left(\frac{\pi}{3}\right)^4}=\frac{\frac{1}{2}}{\frac{\pi^4}{81}}=\frac{81}{2\pi^4}$$

$$\pi^4y\left(\frac{\pi}{3}\right)=\frac{81}{2}$$

11. If coefficients of x, x^2 and x^3 are in arithmetic progression of the binomial expansion of $(1 + x^2)^2 (1 + x)^n$, $n \in N$. Then sum of all values of n is

(1) 9 (2) 7

(3) 8 (4) 10

Answer (1)

Sol. Coefficient of x in

$(1 + 2x^2 + x^4)\ (1 + x)^4$ is

${}^nC_1$

Coefficient of x^2 is

$\Rightarrow {}^nC_2 + 2\cdot{}^nC_0$

Coefficient of x^3 is

${}^nC_3 + 2\cdot{}^nC_1$

$\Rightarrow$ since they are in *A.P.*

$$\Rightarrow \left({}^nC_1\right)+\left({}^nC_3+2\cdot{}^nC_1\right)=2\left({}^nC_2+2\cdot{}^nC_0\right)$$

$$\Rightarrow {}^nC_3+3\cdot n=2.\left({}^nC_2+2\right)$$

$$\Rightarrow \frac{n(n-1)(n-2)}{6}+3n=2(n(n-1)+4$$

$\Rightarrow (n^2–n)\ (n –2) + 18x = 6n^2 – 6n + 24$

$\Rightarrow n^3 – 3x^2 + 20n = 6n^2 – 6n + 24$

$\Rightarrow n^3 – 9n^2 + 26n – 24 = 0$

$n = 2, 3, 4$

$\Rightarrow$ sum of all $n = 2 + 3 + 4 = 9$

12. Let $f(x)=\int\frac{e^x(2-x^2)}{\sqrt{1+x}(1-x)^{\frac{3}{2}}}dx$, with $f(0) = 0$, then $f\left(\frac{1}{2}\right)$ is

(1) $\sqrt{2e}+1$ (2) $\sqrt{2e}-1$

(3) $\sqrt{3e}+1$ (4) $\sqrt{3e}-1$

Answer (4)

Sol. $$f(x)=\int e^x\left[\frac{1}{\sqrt{1-x^2}(1-x)}+\frac{(1-x^2)}{\sqrt{1-x^2}(1-x)}\right]dx$$



$$= \int e^x \left(\frac{1}{(1-x)\sqrt{1-x^2}} + \frac{\sqrt{(1-x^2)}}{\sqrt{(1-x^2)}} \right)$$

$$= \int e^x \left(\underbrace{\frac{1}{(1-x)\sqrt{1-x^2}}}_{f'(x)} + \underbrace{\sqrt{\frac{1+x}{1-x}}}_{f(x)} \right)$$

$$= e^x \sqrt{\frac{1+x}{1-x}} + c$$

$\because f(0) = 0 \Rightarrow 0 = 1 + c \Rightarrow c = -1$

$$f\left(\frac{1}{2}\right) \Rightarrow e^{\frac{1}{2}}\sqrt{3} - 1$$

13. If A is matrix of order 3 and $x = |3\text{adj}(A^2)\cdot\text{adj}(2A)|$ and $|A| = 6$ and $x = 2^n\cdot 3^m$, then $m + n$ is

(1) 21 (2) 25

(3) 27 (4) 19

Answer (1)

Sol. $|3\text{adj}(A^2)\cdot\text{adj}(2A)| = 3^3|\text{adj}(A^2)||\text{adj}(2A)|$

$= 3^3|A^2|^2|2A|^2 = 3^3|A|^4\cdot 2^6|A|^2$

$= 3^3\cdot 2^6\cdot|A|^6 = 3^3\cdot 2^6\cdot 6^6$

$= 2^{12}\cdot 3^9$

$m + n = 21$

14. Let the domain of $f(x) = \log_3\left(\log_5\left(\log_7\left(9x - x^2 - 13\right)\right)\right)$ is (m, n). Let $\frac{n}{3}$ and $\frac{3m}{8}$ be eccentricity and length of latus rectum of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ respectively, then the value of $\left(\frac{a+b^2}{a-b^2}\right)$ is equal to

(1) 7 (2) 9

(3) 3 (4) 13

Answer (1)

Sol. $\Rightarrow \log_5\left(\log_7\left(9x - x^2 - 13\right)\right) > 0 = \log_5 1$

$\log_7\left(9x - x^2 - 13\right) > 1 = \log_7 7$

$\Rightarrow 9x - x^2 - 13 > 7$

$\Rightarrow 9x - x^2 + 20 < 0 \Rightarrow x \in (4, 5)$

$\Rightarrow e = \frac{5}{3}, \frac{2b^2}{a} = \frac{3}{8}\times 4 = \frac{3}{2}$

$\Rightarrow \frac{b^2}{a} = \frac{3}{4} \Rightarrow \frac{a+b^2}{a-b^2} = \frac{3+4}{4-1} = 7$

$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{3a}{4a^2} = \frac{25}{9}$

$\Rightarrow 1 + \frac{3}{4a} = \frac{25}{9} \Rightarrow \frac{30}{4a} = \frac{16}{9}$

$\Rightarrow a = \frac{27}{64}, b^2 = \frac{81}{256}$

15. If point B and C lies on line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$ and point A is (1, 6, 3). If $BC = 10$. Then, the area of ΔABC is

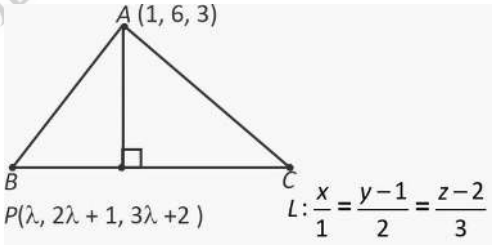
(1) $2\sqrt{13}$ (2) $5\sqrt{13}$

(3) $6\sqrt{13}$ (4) $4\sqrt{13}$

Answer (2)

Sol. $L: \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

Let any point on line L is $(\lambda, 2\lambda + 1, 3\lambda + 2)$



$1(\lambda - 1) + 2(2\lambda - 5) + 3(3\lambda - 1) = 0$

$\lambda - 1 + 4\lambda - 10 + 9\lambda - 3 = 0$

$14\lambda - 14 = 0$

$\lambda = 1$



$P(1, 3, 5)$

$AP = \sqrt{(1-1)^2 + (6-3)^2 + (3-5)^2}$

$= \sqrt{9+4}$

$= \sqrt{13}$

Area $(\Delta ABC) = \frac{1}{2} \times 10 \times \sqrt{13}$

$= 5\sqrt{13}$

16.

17.

18.

19.

20.

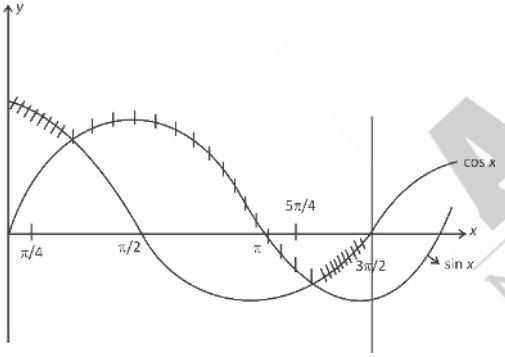
SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. The area (in square units) bounded by the curve $f(x)$ = Max{sin$x$, cos$x$} and x-axis between $x = 0$ and $x = \frac{3\pi}{2}$ is A. Then, the value of $A + A^2$ is

Answer (3)

Sol.



Requiret area $= \int_0^{\pi/4} \cos x\, dx + \int_{\pi/4}^{\pi} \sin x\, dx$

$+ \int_{\pi}^{5\pi/4} (-\sin x)\, dx + \int_{5\pi/4}^{3\pi/2} (-\cos x)\, dx$

$= [\sin x]_0^{\pi/4} + [-\cos x]_{\pi/4}^{\pi} + [\cos x]_{\pi}^{5\pi/4} - [\sin x]_{5\pi/4}^{3\pi/2}$

$= \frac{1}{\sqrt{2}}\left[1 + \frac{1}{\sqrt{2}}\right] + \left[-\frac{1}{\sqrt{2}} + 1\right] - \left[-1 + \frac{1}{\sqrt{2}}\right]$

$= \sqrt{2} + 1 - \frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}}$

$= 3$

22. The mean and variance of the 8 observations –10, –7, –1, x, y, 16, 2, 9 are $\frac{7}{2}$ and $\frac{293}{4}$ respectively. Then, the mean of x, y, $x + y + 1$, $|x - y|$ is

Answer (11)

Sol. Mean = $\frac{7}{2}$

$\frac{-10-7-1+x+y+16+2+9}{8} = \frac{7}{2}$

$\Rightarrow x + y + 9 = 28$

$\Rightarrow \boxed{x + y = 19}$

Var = $\frac{293}{4}$

$\frac{100+49+1+x^2+y^2+256+4+8}{8} - \frac{49}{4} = \frac{293}{4}$

$\frac{491 + x^2 + y^2}{8} = \frac{342}{4} \times 2$

$x^2 + y^2 = 193$

$\Rightarrow (x, y) = (7, 12)$ or $(12, 7)$

Mean = $\frac{x+y+x+y+1+(x-y)}{4}$

$= \frac{19+20+5}{4}$

$= 11$

23.

24.

25.

❑ ❑ ❑

